

Seminar 2: Artificial Intelligence for Software Engineering

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Short Biography

Phuong Nguyen

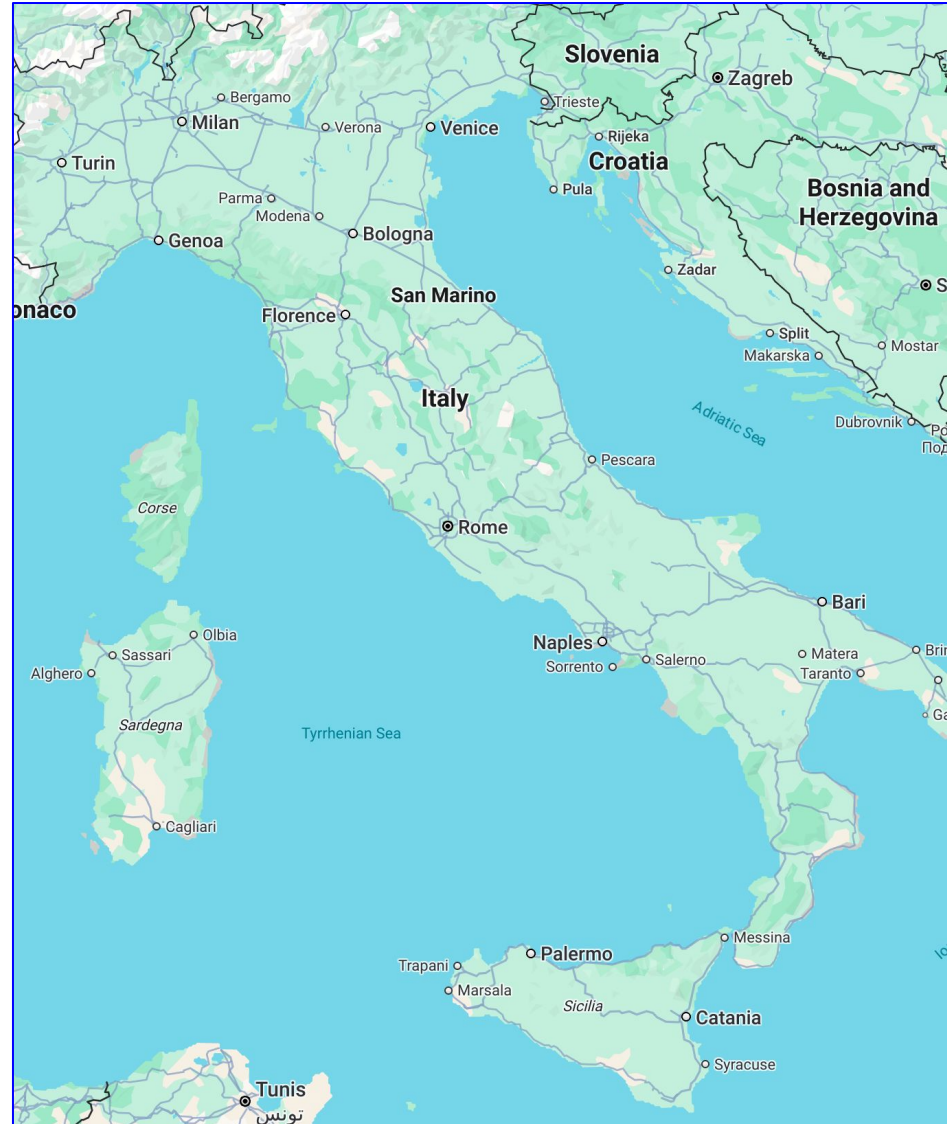
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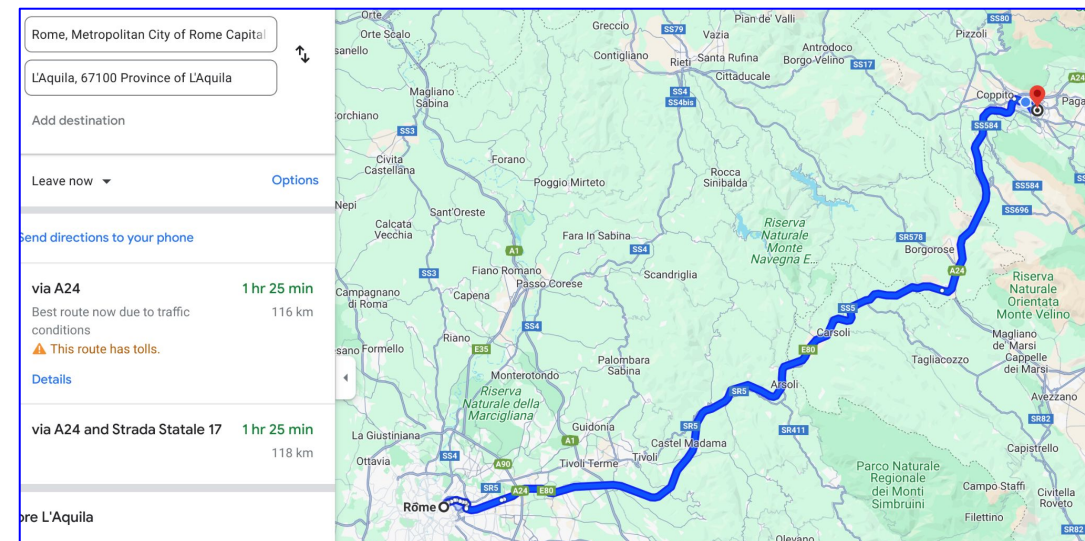
Website: <https://www.disim.univaq.it/ThanhPhuong.html>

- PhD in Computer Science: University of Jena (Germany).
- Lecturer: FPT University, Duy Tan University (Vietnam).
- Postdoctoral researcher: Polytechnic University of Bari, University of L'Aquila (Italy).
- From 02/2022 - 01/2025: Assistant Professor, University of L'Aquila.
- From 02/2025: Associate Professor, University of L'Aquila.
- Research Interests: Machine Learning, Recommender Systems, Software Engineering.

The city of L'Aquila



The city of L'Aquila



- Located in the central part of Italy, about 120 km east of Rome.
- A small, and quiet city.
- Beautiful landscape weather, a bit cold, but not very cold.

The city of L'Aquila



The city of L'Aquila



University of L'Aquila



- Founded in 1596, and 1964 with nine departments, and around 18,000 students
- The University is among the top 900 in 2021 (top 800 in 2020) and top 40 in Italy
- In the subject ranking UnivAQ is listed among:
 - 151-200 in Mathematics (8-12 in Italy).
 - 401-500 in Materials Science and Engineering (9-16 in Italy).
 - 401-500 in Electrical & Electronic Engineering (14-23 in Italy).

- Applications of Artificial Intelligence (AI) in Software Engineering:
 - Basic concepts in AI.
 - Using Machine Learning to mine data from open source platforms.
 - Providing developers with suitable recommendations.

- Scientific Writing
 - Possible topics for projects, papers.
 - How to structure a thesis, a paper.
 - How to write different parts of a thesis/paper.

Agenda

- Basic concepts in Machine Learning.
- Introduction to recommender systems.
- Recommendation of API function calls and code snippets.
- Classification of metamodels using CNNs.
- Detection of technical debt in source code.
- Recommendation of GitHub topics.
- Detection of source code written by AI.
- Categorization of GitHub actions.
- Scientific writing.

Teaching methods

- Lectures, seminars, and code demonstrations.
- The teaching focuses on learning-by-doing, and taking students through realistic projects.
- Extensive use of problem-based learning.
- Students are exposed to sufficient levels of detail and risk, thus gaining an appreciation for the challenges of solving actual data science problems.

Student involvement

- Students are encouraged to participate in the lecture by having questions/comments.
- Students can investigate a specific topic, and present in the class.
- It is highly advised that students use GitHub to store their projects and to collaborate with other team members.

Assignment and Exam

- Project and oral exam.
- There will be a list of projects so that you can choose.
- Projects are related to the application of Machine Learning and Recommender Systems in Software Engineering.
- Reports should be written in English.
- Each group with 2 or 3 students works on a common project.
- Every member of the group should contribute to the project. I am going to check this during the exam (Please be prepared :-).

Publications by SOICT students



CORE Rank A*

SHORT-PAPER | OPEN ACCESS | CC BY



Teamwork makes the dream work: LLMs-Based Agents for GitHub README.MD Summarization

Authors: [Duc S. H. Nguyen](#), [Bach G. Truong](#), [Phuong T. Nguyen](#), [Juri Di Rocco](#), [Davide Di Ruscio](#) | [Authors Info & Claims](#)

FSE Companion '25: Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering
Pages 621 - 625 • <https://doi.org/10.1145/3696630.3728511>

Published: 28 July 2025 [Publication History](#)



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Publications by SOICT students

SCImago Q2

Automated Software Engineering (2026) 33:43
<https://doi.org/10.1007/s10515-025-00588-4>



Automated summarization of software documents: an LLM-based multi-agent approach

Duc S. H. Nguyen¹ · Minh T. Nguyen¹ · Phuong T. Nguyen² · Juri Di Rocco² ·
Davide Di Ruscio²

Received: 4 September 2025 / Accepted: 11 December 2025

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2026

Publications by SOICT students

SCImago Q1

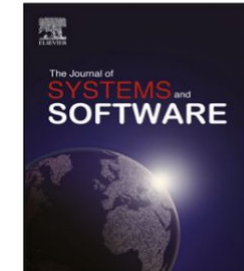
The Journal of Systems and Software 224 (2025) 112375



Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss



ENSESMELLS : Deep ensemble and programming language models for automated code smells detection[☆]

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Publications by SOICT students

EASE 2025

Tue 17 - Fri 20 June 2025 Istanbul, Turkey

CORE Rank A

Bake Two Cakes with One Oven: RL for Defusing Popularity Bias and Cold-start in Third-Party Library Recommendations

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Publications by SOICT students



Publications by SOICT students

EASE 2023

Tue 13 - Fri 16 June 2023 Oulu, Finland

CORE Rank A

Fusion of deep convolutional and LSTM recurrent neural networks for automated detection of code smells

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— Publications by SOICT students

ESEIW 2025

Mon 29 September - Fri 3 October 2025

CORE Rank A

When Retriever Meets Generator: A Joint Model for Code Comment Generation

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Publications by SOICT students

SCImago Q1



Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss



Walking on thin ice? Defending RAG-based code recommender systems against Poisoning attacks[☆]

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Massimiliano Di Penta^c

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Grading

Task	Mark (%)
Selecting and adopting suitable techniques	20
Designing and implementing the application components	40
Performing evaluation of the application	20
Understanding the knowledge of the methodologies and technologies presented during the course	10
Participation (interaction) in the lectures	10

Libraries and platforms

matplotlib

Pandas



NumPy

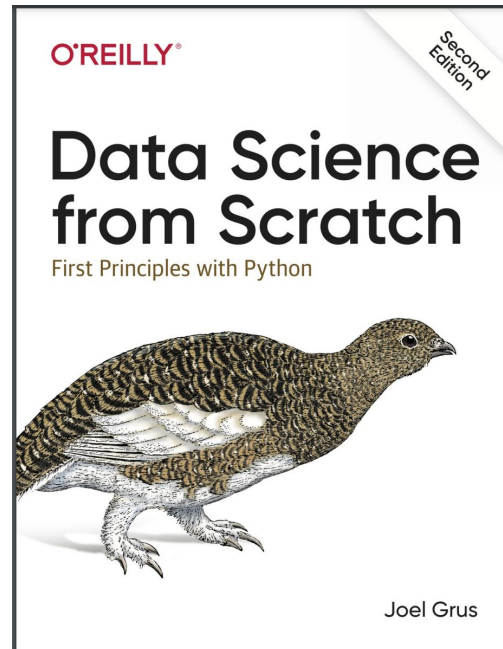


seaborn

colab

- There are various libraries in Python, ready to be used for different tasks
- Students are encouraged to use these libraries/platforms for their projects

References



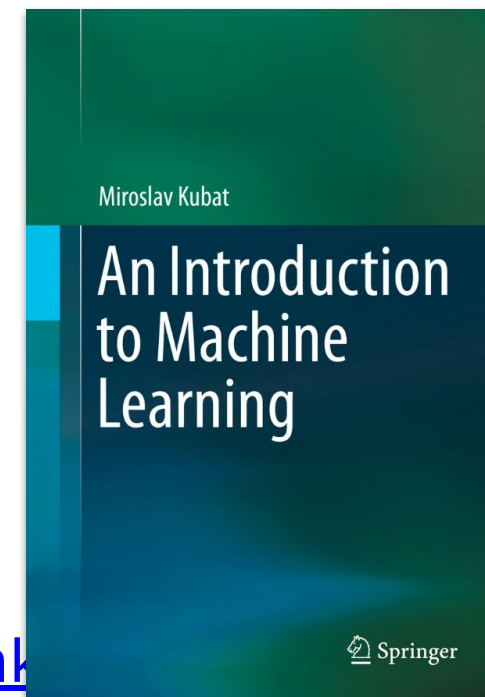
- Joel Grus, “***Data Science from Scratch***” ([Link](#)).
- Alberto Boschetti and Luca Massaron, “***Python Data Science Essentials***” ([Link](#)).
- Datasets for working with data ([Link](#)).
- Kaggle: A platform for with several Machine Learning problems.

References

Neural Networks and Deep Learning

Michael Nielsen

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<http://neuralnetworksanddeeplearning.com>



- Michael Nielsen, Neural Networks and Deep Learning ([Link](#))
- Miroslav Kubat, An Introduction to Machine Learning, DOI: [10.1007/978-3-319-20010-1](https://doi.org/10.1007/978-3-319-20010-1)
- Datasets for testing: <https://archive.ics.uci.edu/>
- Kaggle: A platform for working with several ML.